

Group Project list

1. London Sport 1

Exploring innovative use of OpenActive data to increase physical activity in London

OpenActive is a national platform that makes data on physical activity sessions openly available. The dataset has expanded significantly—from approximately 76,000 opportunities in 2017 to more than 2 million in 2024. Innovators across the sector use this data to develop services such as:

- Apps that help individuals find yoga classes or sports clubs.
- Real-time availability checkers for facilities such as badminton courts.
- Tools that connect people to activities in their local communities.

Through collaboration with a range of data aggregators, OpenActive enables more consistent, accessible and consolidated information for the public. This creates potential to reach new audiences, improve user experience and ultimately make it easier for people to become active.

London Sport aims to strengthen its use of innovative data and digital approaches to support higher physical activity levels across the capital. At present, the system requires new solutions that improve audience engagement and help embed digital pathways into the everyday experience of finding and participating in physical activity.

Students are invited to review, analyse and innovate using OpenActive data, drawing on approaches from across the country and developing new models, prototypes or recommendations tailored specifically to London's diverse communities.

2. London Sport 2

Using machine learning to forecast physical activity participation, volunteering and activity patterns in London

London Sport seeks to strengthen its forecasting capabilities by building a clearer understanding of future scenarios that may affect Londoners' participation in sport, physical activity and volunteering. We are particularly interested in how demographic, environmental and behavioural factors interact across London's boroughs.

Using machine learning techniques, we invite students to explore correlations between demographic characteristics, activity levels (e.g., proportion of individuals undertaking 150+ minutes of physical activity per week), volunteering trends and types of sport participation using data from Sport England's Active Lives Survey. The final outputs should support London Sport in anticipating likely patterns over the next ten years.

The analysis should address the following areas:

1. Overall participation forecasts

- The likelihood of Londoners participating in specific activities over the next decade.
- Projected levels of physical activity across London as a whole over the next ten years.
- Projected levels of sport-specific club membership across London as a whole over the next ten years.
- Forecasts of activity levels for each London borough.
- Key factors that may reduce individuals' available free time and subsequently affect participation in physical activity.

2. Demographic-specific insights

- Projected likelihood of participation in specific activities within each London borough.
- Forecasted participation levels for different demographic groups, including age, gender, ethnicity, disability status and socio-economic groups, and the types of sport or physical activity they are most likely to engage in.
- Likelihood of participation in physical activity among people with general or long-term health conditions.

3. Indoor vs outdoor activity

- The expected balance of indoor versus outdoor physical activity over the next ten years.
- Differences in indoor and outdoor activity preferences among people with varying activity levels (from highly active to least active).
- Relationships between inner vs outer London locations, activity levels and socio-economic factors.

4. Volunteering in sport

- The likelihood of individuals from diverse backgrounds volunteering in sport over the next decade.
- Forecasts of the types of volunteer roles different demographic groups (including ethnicity, disability status, gender and age) are likely to take on.

3. Uk Antarctic Heritage Trust

Antarctic A/V Media enhancement and interrogation

The student project would be working with digital assets from our Media library, primarily images, based around image recognition if feasible. We have decades of visitor photos, condition record images, and artefact images, some video and a little audio from multiple sites.

Project aim:

As our media library is not well organised - we would be open to the use of visual machine learning for image recognition. With the objective to be able to automatically apply classification and categories to the metadata. If successful this would help us organise, label and describe the digital assets to improve searchability, help us to be more efficient, better storage management, open new project opportunities etc.

We can provide a variety of images of our several sites, landscapes and artefacts for use. These images will be in a range of formats.

The above would be the foundation to the below future (or extended) project opportunities and benefits:

- The ability to use images effectively in conservation management and analysis – monitoring building condition or landscape (snow level) changes over time, identify and analyse visitor images to extract information about movements on site and objects of greatest interest.
- A system which can visually analyse and describe an image might enable us to undertake a citizen science project similar to the Happy Whale identification project <https://happywhale.com/home> for our

sites

- Depending on complexity (and agreement from our partners at British Antarctic Survey Archives) the image data could be analysed in conjunction with the recorded staff interviews and transcriptions creating some kind of text question with verbal and visual response based on key words
- We could use the output descriptive and classification data for a variety of levels of analysis and assessment which can then be input into data-driven visualisations based on our media library content

4. IRBM

Integrating Multimodal Spatial Data to Predict Molecular Patterns in Complex Systems

IRBM is a leading research organization with extensive experience in transforming early-stage science into advanced drug discovery programs. The company specializes in cancer, neurological disorders, infectious diseases, and rare diseases. One of IRBM's most active research areas is the identification and characterization of potential therapeutic targets for the treatment of oncologic diseases.

In this context, recent advances in spatial transcriptomics and spatial proteomics have opened new possibilities for understanding how gene expression and protein abundance of potential targets are organized within intact tissue architecture. While spatial transcriptomics provides high-resolution maps of RNA expression, spatial proteomics captures the actual functional molecules operating within cells. Integrating these two complementary modalities offers a powerful opportunity to uncover how transcriptional programs translate into spatially resolved protein patterns, enabling more accurate characterization of tissue microenvironments, cell-cell interactions, and disease-associated molecular states. However, despite this potential, the extent to which spatial proteomic signals reliably mirror spatial transcriptomic expression profiles remains poorly understood and largely unexplored.

This project aims to address that gap by developing an integrative computational framework to:

- a) evaluate the degree of correlation between spatial transcriptomic and spatial proteomic analytic platforms
- b) develop a predictive model capable of inferring spatial proteomic expression patterns from spatial transcriptomic data.

Students will perform a comparative analysis of matched spatial transcriptomic and spatial proteomic datasets to quantify RNA-protein concordance for 44 markers across diverse tissue regions. Building on these insights, the project will explore machine learning and statistical modeling approaches—such as multivariate regression, graph-based methods, or deep learning architectures—to infer protein distribution while preserving spatial context.

The final outcome will be a predictive model capable of reconstructing spatial proteomic landscapes from transcriptomic measurements, accompanied by a thorough evaluation of its performance, biological interpretability, and potential limitations. This work will contribute to the emerging field of multimodal spatial omics integration and help clarify how transcriptional information translates into protein-level organization within complex tissues. The results may provide novel insights that support and enhance drug discovery and target characterization activities carried out by the industrial partner.

5. Swindon Borough Council

Designing inclusive growth: Using applied data science to shape Swindon's economic future

How can applied data science help inform Swindon Borough Council decisions to deliver on our inclusive growth strategy that aims to increase GVA by ~30% by 2036, by strengthening priority sectors and enterprise types?

Swindon Borough Council is developing a new economic strategy focused on driving inclusive growth, and strengthening Swindon's position as a high-growth hub. The council has already an initial algorithmic exploration using open economic data and governed local authority datasets to understand place-based economic structures, sectoral dynamics, and neighbourhood-level variation in outcomes.

This project will build on that foundation and offer MSc Data Science students the opportunity to apply advanced analytical methods to a real public-sector challenge. Students will work with multi-source dataset including open economic indicators (e.g. GVA, skills, enterprise structure, deprivation, connectivity) alongside appropriately governed business rates data, to generate actionable insight that can directly inform policy and investment decisions.

The core aim is not to optimise prediction accuracy, but to use modern data science techniques to deepen understanding of how local economic systems function, how different types of places and sectors contribute to growth, and where targeted intervention may have the greatest impact. Students will be encouraged to explore approaches such as clustering extensions, sector-specific analysis, SME growth modelling, scenario simulation, and interpretable machine learning, and to bring their own methodological perspectives to the problem.

6. AstraZeneca

Building a Recommendation System for Cell Line Selection Using Integration of Biological Datasets

Cell lines are foundational to the development, manufacturing, and quality control of biopharmaceuticals at AstraZeneca. You will build a tool called CellLineSelector which will help AstraZeneca to select cell lines.

CellLineSelector is an tool that recommends the most suitable cell lines for a given experiment by integrating and interpreting publicly available expression datasets across modalities (bulk/single-cell RNA-seq, proteomics, CRISPR screens). It provides transparent, context-aware rationales based on pathway activity, lineage, mutations, receptor/marker status, and assay compatibility, while flagging trade-offs and data gaps. The result is faster, reproducible, and biologically grounded cell line selection with auditable evidence.

The platform will unify publicly available transcriptomic and proteomic datasets (DepMap, Human Protein Atlas, GEO) to identify human cell lines that express one or more user-defined target genes, while also enabling the exclusion of lines expressing undesirable or confounding genes. Through the ingestion, harmonization, and indexing of multi-omic evidence, CellLineSelector will address dataset sparsity and heterogeneity to deliver ranked cell-line recommendations accompanied by transparent, interpretable confidence scores. In addition, the system will compute similarity metrics across

recommended cell lines, leveraging the full breadth of integrated data to support robust, evidence-based experimental planning.

CellLineSelector will support two complementary analytical frameworks, each contributing distinct strengths to cell-line prioritization and justification:

1. **Classical Statistical Model** This framework will apply rigorous data-aggregation strategies, missing-data modelling, thresholding, and quantitative confidence scoring to generate reproducible and transparent evidence-based rankings of candidate cell lines.
2. **Agentic AI Model** This model will leverage retrieval-augmented reasoning to prioritize cell lines and produce structured, interpretable justifications, enabling users to understand the underlying evidence supporting each recommendation.

Objectives:

Integrate multi-omics public datasets for human cell lines into a unified, queryable knowledge graph to enable coherent cross-dataset analyses.

Enable target-driven cell-line selection based on the presence or absence of gene expression across user-defined molecular targets.

Quantify confidence levels despite data sparsity, variability across studies, and heterogeneous measurement platforms.

Provide transparent, reproducible justifications for all recommendations to support defensible experimental planning.

Deliver both programmatic (API) and user-friendly (UI) interfaces, including exportable reports for downstream documentation and review.

7. Lloyds Bank

Identifying Customer needs via Companies House & Unstructured Data

We want to explore whether publicly available data can help identify potential business opportunities or risks for clients and potential clients. Specifically, the goal is to link structured company data from Companies House with unstructured media data (e.g., news articles, global event databases) to detect patterns in external behaviour and news coverage. These patterns could indicate a company's future needs, such as lending, business support, or growth opportunities, and help the bank proactively engage with clients or prospects who are not currently banking with Lloyds.

In short:

Can external public data (company filings + media coverage) be used to predict business needs and inform commercial banking strategies?

8. McCain Foods + Dalhousie University

On-the-go detection of field rocks for precision agriculture in potato fields

This is a joint project between Dalhousie University and McCain Foods Limited (Canada) to detect rocks in potato fields. The identification of the presence of rocks in the field, the estimation of their size, and the flagging of whether they are partially buried, will help the stakeholder to determine the resources needed to remove rocks in later stages of cultivation. The objective of this project is to develop a machine vision system able to detect rocks lying in fields with different backgrounds. The project will require using machine learning to segment and detect the rocks, as well as estimating their size. Models are expected to be able to run on edge devices, so teams are expected to focus on lighter models if possible, and specify the hardware and software requirements to accomplish this detection in a timely manner, assuming that the machine vision system would be mounted on a potato planter to map the locations of rocks in fields.